Zilionis 2019 NSCLC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

6 samples (≥131-cell floor) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached; iLISI 3.42/6; same-sample neighbour fraction 0.91→0.40 across the θ sweep). Roundish body (UMAP aspect 1.13), single connected component.

Gate 2 — module geography → NO-GO

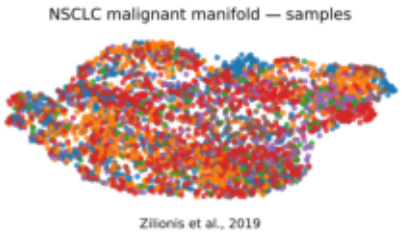
Embryonic pattern absent. In score space the modules look anti-correlated (Gast/pSom −0.53, ≈ Puram's branching −0.49) — but this is a shared-control-pool artifact: on RAW mean-expression all three POSITIVELY co-activate (Gast/pSom +0.58, Gast/Crest +0.45, pSom/Crest +0.51; survives library-size regression). One shared developmental field, not competing arms. No convergence hub (all joint-high enrichments depleted 0.07–0.62); Crest not terminal (high-decile pt 0.56, mid-axis); every principal-tree tip is pSom-dominant (no Gast/Crest arm).

Gate 3 — cross-repression → NA

Not branching and Gate 2 NO-GO — no branch pair / hub to test for mutual exclusivity.

STEP FIGURES

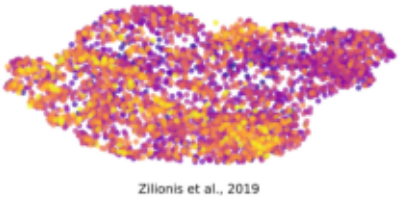
Manifold — samples



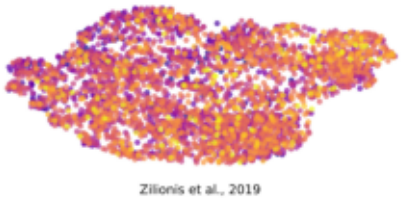
Pseudotime



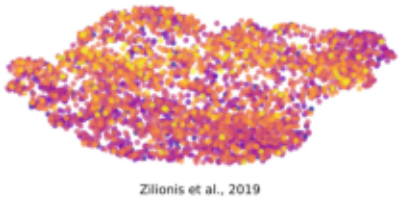
Gastrulation



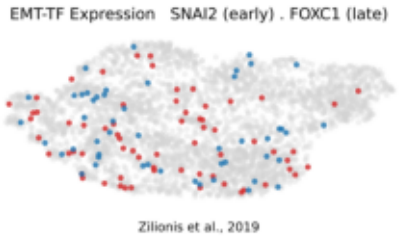
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
NSCLC (Zilionis et al., 2019), inDrop (droplet UMI) — 7 patients / 3,995 malignant cells; 3,920 across 6 samples (all clear the ≥131 floor) enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict.
Embryonic pattern does NOT reproduce — Gate 1 GO / Gate 2 NO-GO / Gate 3 NA. The malignant manifold is a single continuous, sample-integrated cloud organized by carcinoma metaprograms (Alveolar / cell-cycle / Stress / IFN; Moran's I 0.54 / 0.24 / 0.22 vs the embryonic modules' 0.10–0.26), not by an embryonic hub-and-arms axis. The three modules form one shared low-amplitude activation field (raw co-activation +0.45–+0.58): no Gast/pSom co-expression hub, no Gast or Crest arm (all tree tips pSom-dominant), and Crest sits mid-pseudotime (0.56), not terminal — the Peng2019 / Hwang2022 NO-GO signature, opposite Steele 2020's terminal-Crest GO. The score-space −0.53 that read as branching in five prior attempts is a union-control-pool subtraction artifact, not cross-repression. EMT-TF step run unconditionally: SNAI2 (early) → FOXC1 (late) chosen by data over Puram's SNAI1/ZEB1, all near the detection floor with negligible end-separation (Δpt ≈ 0) — consistent with no EMT axis.

Parameter choices

Min cells/sample	≥131 (gap 232→75, r3.09; drops p2)	**Integration**	Harmony θ=10; 20 PCs; spread 1.3
Scope	whole manifold (no scoping)	**Trajectory**	curve N=20, λ=0.3, μ=100; fold-free 2.46
Root node	Gast-high, Crest-low tip (origin)	**Overdispersed**	248 genes; PPT tree rejected
Geometry	Non-branching; embryonic pattern absent → NO-GO	**EMT-TFs**	SNAI2 (early) → FOXC1 (late); near floor, Δpt≈0